



Sir Syed University of Engineering & Technology

END SEMESTER EXAMINATIONS FALL 2022

Faculty	Faculty of Basic & Applied Sciences	Department	Computer Science	Batch	2021F
Semester	3 rd	Degree Program	BS(CS)	Course Instructor	Fariha Ansari/Maria Rashid
Course Code	MS104	Course Title	Differential Equations	Credit Hours	3+0
Date of Examinations	15 th Feb 2023	Timings	9:30am-12:30pm	Shift	Morning

Timings: 3 Hours

Max Marks: 50

Instructions:

- Please don't write anything on the Question Paper except for your Roll No. and Name.
- The paper must be solved / attempted on Printed Answer books provided by the Examinations Department.
- Please return the Question Paper along with the Answer books.

Roll No.		Name		Section		'A' Copy Serial No.	
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Q.No.1	(12 Marks)
a) Solve the given first order differential equations:	[04+04 marks]

i) $y' \cos y + x \sin y = 2x$

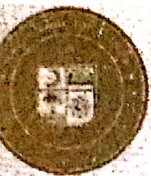
ii) $5dx - e^{y-x} dy = 0$

b) Find the Orthogonal Trajectory of the curve $y = \sqrt{x+c}$ [04 marks]

Q.No.2	(10 Marks)
Solve the given non homogeneous second order differential equations:	[05+05 Marks]

i) $y'' - 2y' + y = e^x \sin x$

ii) $(x^2 D^2 - 2)y = 9x^2$



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Q.No.3

(14 Marks)

Evaluate the Fourier Series of the given functions:

[07+07 Marks]

i) $f(x) = x + |x|$; $(0 \leq x \leq 2\pi)$

ii) $f(x) = x^3$; $(-1 \leq x \leq 1)$

Q.No.4

(14 Marks)

Solve the following Partial Differential Equations:

[07+07 Marks]

i) $yu_{xy} - xu = 0$

ii) $u_{xy} - u_{yy} = 0$ ($v = x$ and $z = x + y$)